

5.6 Classwork: “Show That” Problems — Markscheme

1. [Maximum mark: 16]

A particle P moves along the x -axis. The velocity of P is $v \text{ ms}^{-1}$ at time t seconds, where

$$v(t) = 4 + 4t - 3t^2 \quad \text{for } 0 \leq t \leq 3.$$

When $t = 0$, P is at the origin O .

- (a) (i) Find the value of t when P reaches its maximum velocity. [7]

Maximum velocity when $v'(t) = 0$ (M1)

$$v'(t) = 4 - 6t = 0$$

$$t = \frac{2}{3} \quad (\text{A1})$$

- (ii) Show that the distance of P from O at this time is $\frac{88}{27}$ metres.

$$s(t) = \int v \, dt = 4t + 2t^2 - t^3 (+c) \quad (\text{M1})$$

$$s(0) = 0 \Rightarrow c = 0 \quad (\text{A1})(\text{A1})$$

$$s\left(\frac{2}{3}\right) = 4\left(\frac{2}{3}\right) + 2\left(\frac{2}{3}\right)^2 - \left(\frac{2}{3}\right)^3 \quad (\text{M1})$$

$$= \frac{8}{3} + \frac{8}{9} - \frac{8}{27} \quad (\text{A1}) \quad (\text{AG})$$

$$= \frac{72 + 24 - 8}{27} = \frac{88}{27}$$

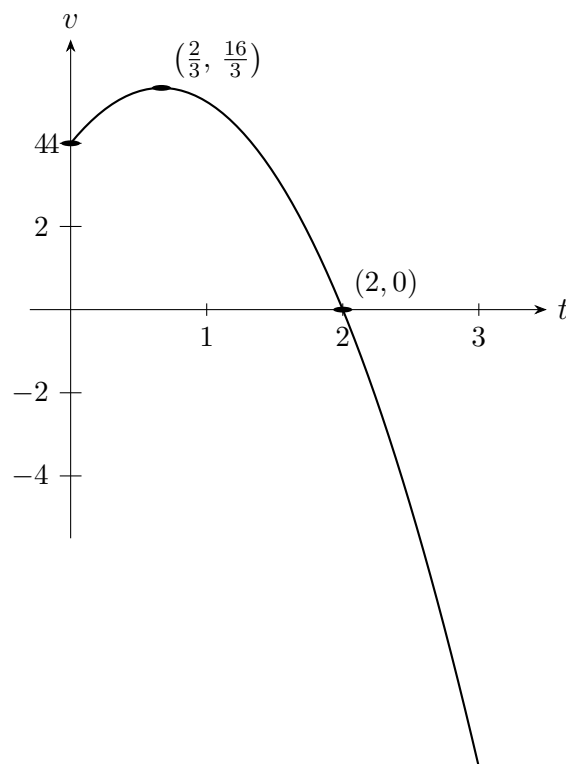
- (b) Sketch a graph of v against t , clearly showing any points of intersection with the axes. [4]

$$v(0) = 4 \text{ (intercept)} \quad (\text{A1})$$

$$v(t) = 0: 3t^2 - 4t - 4 = 0$$

$$(3t + 2)(t - 2) = 0 \Rightarrow t = 2 \quad (t > 0) \quad (\text{A1})$$

$$\text{Maximum at } t = \frac{2}{3}: v\left(\frac{2}{3}\right) = 4 + \frac{8}{3} - \frac{4}{3} = \frac{16}{3} \quad (\text{A1})$$



(A1) shape

(c) Find the total distance travelled by P .

[5]

Recognize v changes sign at $t = 2$

(M1)

$$\int_0^2 v \, dt = [4t + 2t^2 - t^3]_0^2$$

(M1)

$$= 8 + 8 - 8 = 8$$

(A1)

$$\int_2^3 v \, dt = [4t + 2t^2 - t^3]_2^3$$

(A1)

$$= (12 + 18 - 27) - 8 = -5$$

(A1)

$$\text{Total distance} = 8 + |-5| = 13 \text{ m}$$

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2. [Maximum mark: 13]

Let $y = \frac{\ln x}{x^4}$ for $x > 0$.

(a) Show that $\frac{dy}{dx} = \frac{1 - 4 \ln x}{x^5}$. [3]

$$\text{Quotient rule: } \frac{dy}{dx} = \frac{x^4 \cdot \frac{1}{x} - \ln x \cdot 4x^3}{(x^4)^2} \quad (\text{M1})$$

$$= \frac{x^3 - 4x^3 \ln x}{x^8} \quad (\text{A1})$$

$$= \frac{x^3(1 - 4 \ln x)}{x^8} = \frac{1 - 4 \ln x}{x^5} \quad (\text{A1}) (\text{AG})$$

(b) The graph of f has a horizontal tangent at point P . Find the coordinates of P . [5]

$$f'(x) = 0 \Rightarrow 1 - 4 \ln x = 0 \quad (\text{M1})$$

$$\ln x = \frac{1}{4} \quad (\text{A1})$$

$$x = e^{1/4} \quad (\text{A1})$$

$$y = \frac{\ln e^{1/4}}{(e^{1/4})^4} = \frac{\frac{1}{4}}{e} = \frac{1}{4e} \quad (\text{M1})$$

$$P\left(e^{1/4}, \frac{1}{4e}\right) \quad (\text{A1})$$

(c) Given that $f''(x) = \frac{20 \ln x - 9}{x^6}$, show that P is a local maximum point. [3]

$$f''(e^{1/4}) = \frac{20 \cdot \frac{1}{4} - 9}{(e^{1/4})^6} = \frac{5 - 9}{e^{3/2}} = \frac{-4}{e^{3/2}} \quad (\text{M1})$$

$$f''(e^{1/4}) < 0 \quad (\text{A1})$$

Therefore P is a local maximum. (R1) (AG)

(d) Solve $f(x) > 0$ for $x > 0$. [2]

$$\frac{\ln x}{x^4} > 0 \Rightarrow \ln x > 0 \quad (\text{since } x^4 > 0) \quad (\text{A1})$$

$$x > 1 \quad (\text{A1})$$

Adapted from IB AA SL 2021 May P1 TZ1 Q8